

Programming Using C++ Level 1

For Loops & Arrays



Programming Using C++

Level 1

For Loops & Arrays

Definition

- ◆ A for loop can be combined with arrays, structures, and functions to process multiple records efficiently.
- ◆ An array stores multiple structure objects, while functions can read and display each object.
- ◆ In this program, the user can enter information for N persons and then display all of them.

Explanation

◆ The strInfo structure stores:

 FirstName

 LastName

 Age

 Phone

◆ Persons[100] is an array that can store up to 100 strInfo objects.

◆ ReadPersonsInfo() uses a for loop to read information for each person.

◆ PrintPersonsInfo() uses another for loop to display the information for each person.

◆ ReadInfo() and PrintInfo() handle the information of one person at a time.

☐ Steps in main()

☐ Define the structure

◆ Create strInfo with FirstName, LastName, Age, and Phone.

☐ Create the array

◆ Create Persons[100] to store information for multiple persons.

☐ Read the number of persons

◆ Store the user's number in NumberOfPersons.

☐ Start the first loop

◆ Start i at 0 and continue while $i \leq \text{NumberOfPersons} - 1$.

☐ Read each person's information

◆ Pass Persons[i] to ReadInfo().

☐ Start the second loop

◆ Loop through all stored persons again.

☐ Print each person's information

◆ Pass Persons[i] to PrintInfo().

☐ Display all records

◆ Continue until all NumberOfPersons records have been processed.

💡 Example

◆ If the user enters:

Number of persons = 2

◆ The program stores:

Persons[0] → First person's information

Persons[1] → Second person's information

◆ The loop accesses each person using:

Persons[i]

◆ For the first iteration:

i = 0

◆ For the second iteration:

i = 1

➡ Result

◆ The program first asks for the information of each person.

◆ After all information is entered, it displays the information for every person in the array.

Note

- ◆ Array indexes start from 0.
- ◆ If NumberOfPersons is 3, the valid indexes used are:
 - 0
 - 1
 - 2
- ◆ The condition $i \leq \text{NumberOfPersons} - 1$ is equivalent to $i < \text{NumberOfPersons}$.
- ◆ ReadInfo() receives the structure by reference, so it can modify the original structure stored in the array.
- ◆ PrintInfo() receives the structure by value, so it works with a copy when displaying the information.

⊘ Warning

- ◆ The array has a fixed size of 100, so NumberOfPersons should not be greater than 100.
- ◆ Using $i < \text{NumberOfPersons}$ is usually clearer than $i \leq \text{NumberOfPersons} - 1$.



Comparison

▨ Using for Loop with an Array

- ✓ Processes many structure objects automatically.
- ✓ Avoids repeating the same code for every person.
- ✓ Works well when the number of records can change.

▤ Without a Loop

- ✓ Each person would need separate code.
- ✓ The program would become much longer.
- ✓ Adding more persons would require additional code.

✦✦ Summary

- ✓ Structures organize the information for one person.
- ✓ Arrays store information for multiple persons.
- ✓ For loops and functions allow the program to process all persons efficiently.

Thank you 😊